

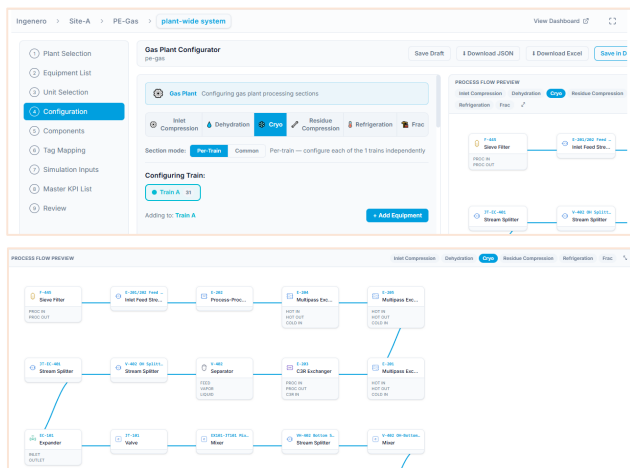
PRODUCT OVERVIEW

GasX360 is a decision-support application for gas processing and NGL recovery plants. Domain and first-principles/ML models are configured on the plant's own historian and DCS tag data, covering feed variability, inlet compression and refrigeration performance, cryogenic train performance, ethane recovery/rejection, propane recovery, frac optimization, slug catcher/inlet separation management and overall unit energy consumption. Outputs reach the process engineer as ranked, quantified recommendations against the parameter that caused them and the constraint they respect. Every model is configured, validated and commissioned per site by Ingenero process engineers.



01 Unit Configuration

Shown: Representative gas plant configuration.



Asset Hierarchy

PIG Receiver, Slug Catcher, Inlet Compression, Dehydration, Cryo Unit, Frac Train

Refrigeration Loops

Propane Refrigeration Loop, Turboexpander Loop, Reflux Loop

Operating Modes

Normal Production, Turndown, KPI reevaluation per asset



02 Plant Data & Model Execution

Shown: Plant-data path to the executive KPI view.



Inputs

Feed Gas Flow, Composition (GC), Ambient conditions, Pressure/Temperature sensor data.

Processing

Tag Validation, Feed reconciliation, Steady state framing and cryo/frac cycle segmentation

Outputs

Performance KPIs, optimal set-points, actionable real-time trends and deviation alerts



Contextualize



Calibrate



Analyze



Recommend



Act



03 AI-Powered Model Agent Suite

Operating constraints that leave value on the table — and the module that recovers it.



Recovery/Rejection Optimization — Dynamic adjustment of the operating set-points/splits to optimize Recovery/Rejection operation through real-time calibrated models.



Slug Catcher Overload — Pigging runs & flow-regime slugs swing separator levels, risking liquid carryover and compressor trips



Inlet/Refrigeration/Residue Gas Compression — Keeps each compression train off its minimum-power floor and clear of surge in real time



Cryo Exchanger Performance Improvement — Tracks the exchanger performance through UA, Approach, maldistribution through temperature profile and early detection of excursions using outlier detection



Multi Cryo/Frac Units Performance Optimization — Improved overall margins in cryo and frac optimization through optimum feed allocation



Moisture Carryover & Hydrate Risk — Dehydration bed breakthrough drives ice/hydrate formation and temperature excursions in the cold box, caught early through outlier detection



Off-Spec NGL Products — Demethanizer/Deethanizer/Depropanizer/Debutanizer reflux and reboiler duty miss purity spec, Excess energy consumption



Compressor Performance Degradation — Outlier detection for tracking compressor/expander reliability, Tracks head, efficiency and surge margin drift over time to catch degradation before it limits capacity



Controller Loop Monitoring with Close Loop Integration — Tracks advanced regulatory and multivariable controller performance & closes the loop by writing optimized set-points through the historian for real-time action.



Unit Energy Minimization — Targets total plant specific energy consumption against throughput, varying feed conditions and ambient conditions



04 Opportunity and Operator Actions

ACTIONABLES — GAS PLANT CRYO UNIT RECOMMENDATION				
SYSTEM	CAUSE MESSAGE	ACTUAL	OPTIMUM	SUGGESTION
TURBOEXPANDER PERFORMANCE	EXPANDER INLET TEMPERATURE HIGH	-82.4 °F	-95.0 °F	INCREASE PRE-COOLING DUTY TO BRING TURBOEXPANDER INLET TEMPERATURE DOWN TO THE OPTIMUM SETPOINT OF -95°F. THIS WILL IMPROVE C2+ RECOVERY
	DEMETHANIZER OVERHEAD TEMPERATURE OVERRUN	-140.2 °F	-148.5 °F	REDUCE DEMETHANIZER OVERHEAD TEMPERATURE TO OPTIMUM SETPOINT BY TRIMMING REFLUX AND REBOILER DUTY.
DEMETHANIZER AND CRYO EXCHANGERS	COLD BOX APPROACH TEMPERATURE OVERRUN	7.20 °F	4.50 °F	OPTIMIZE HP STREAM DEMAND AND HEAT EXCHANGE. EVENTUALLY THIS WILL BRING COLD BOX APPROACH TEMPERATURE TO OPTIMUM AND LOWER PROPANE REFRIGERATION LOAD. REFER OPTIMIZATION RESULTS FOR DETAILS.

PERFORMANCE KPIs		ACTUAL	OPTIMIZED
TOTAL PLANT FEED	470 470 MMSCFD	ETHANE RECOVERED	22,068 24,432 BPD
PROPANE RECOVERED	10,851 10,890 BPD	OVERALL PROPANE RECOVERY	97.55 97.90%
OVERALL ETHANE RECOVERY	63.97 70.82%	NET PRODUCT VALUE	\$212.2M \$223.0M /YR

Up to
\$20k/day

Improved Margin

By shifting product yield toward the higher-value stream within spec and equipment constraints

2-3%

Recovery Maximization

Through real-time optimization of operating set points

4-5%

Energy Savings

Optimized compressor loading and refrigerant circulation reduce specific power consumption

Basis: Illustrative reference gas plant (Turboexpander NGL recovery train); values are for demonstration only and subject to equipment and product-quality constraints. Screenshots and stat figures are drawn from separate sample cases



05 Platform



Historian connectivity

Plug and play with leading plant historians



Simulation integration

Third party simulators integrate without custom work



Daily reporting

Downloadable daily reports with interactive trends



Gen AI assistant

Answers plant questions in the operating workflow



Build your own dashboard

Engineers assemble the views their plant needs

DELIVERY

Configured, validated and commissioned by Ingenero process engineers.

FLEXIBILITY

Any gas plant configuration — cryo, frac or NGL recovery; selectable KPI set

SCALE

Repeat plants reuse the configuration, with no/minimal rework.

Request a customized demo
marketing@ingenero.com